

IndicXTREME: Dataset Preparation & Benchmarking

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Why Indic languages need a dedicated benchmark

Multilingual models promise zero-shot transfer from high- to low-resource languages. In practice, Doddapaneni et al. (ACL 2023) identify three reasons this breaks down for the 22 languages of the Indian constitution:

1. POOR MODEL COVERAGE

Of the 22 scheduled languages, only 15 are supported by XLM-R — languages like Bodo, Dogri, and Kashmiri are simply missing from corpora like CC-100 and mC4.

2. EVEN "SUPPORTED" LANGUAGES ARE STARVED

Where pretraining data does exist for a low-resource language, it is far smaller than for English — the language gets a poor share of model capacity and vocabulary.

3. POOR BENCHMARK COVERAGE

In XTREME-R, only 3 of 10 tasks cover more than two Indic languages, and no task covers more than seven. 15 of the 22 constitutional languages appear in *no* task at all.

IndicXTREME against the existing landscape

	XTREME	XTREME-R	XGLUE	IndicX-TREME
#Indic lang. tasks	25	28	5	105
Avg. #test ins./task	1691.9	1842.7	3845.6	2008
	Wikipedia	CC-100	mC4	IndicCorp
#Indic lang.	20	15	12	23
#Indic lang. tokens	0.2B	5.0B	20.2B ¹	14.4B
Verified source URLs	✓	✗	✗	✓
	mBERT	XLM-R	MuRIL	IndicBERT
#Indic / #Total langs.	11/104	15/110	16/17	23/24
Fertility (↓)	2.8	2.2	1.7	1.7

Every column tells the same story: existing multilingual resources touch Indic languages only incidentally. IndicCorp is the only pretraining corpus built exclusively from **human-verified source URLs**; IndicXTREME has by far the most evaluation sets (105) of any benchmark; and IndicBERT is the only model supporting nearly all (23/24) of the target languages.

Scope: three contributions, one goal

The paper targets the 22 languages of the 8th schedule of the Indian constitution, spanning 4 language families and spoken by over a billion people (8 among the top-20 most-spoken languages globally). It contributes along three axes:

IndicCorp v2

20.9B-token monolingual pretraining corpus, 24 languages, 4 families — this lecture's focus for **dataset preparation**.

IndicXTREME

9-task, 20-language, human-supervised evaluation benchmark — this lecture's focus for **annotation & benchmarking**.

IndicBERT v2

278M-parameter BERT pretrained on IndicCorp v2, evaluated on IndicXTREME — the model used to demonstrate the benchmark works.

Not all Indic languages are equally resourced

Every language in IndicCorp is tagged with its class from Joshi et al.'s (2020) inclusion taxonomy — 5 is a resource-rich **"Winner,"** 1 is an almost entirely unresourced **"Left-Behind."** Of the languages newly added to IndicXTREME, nine are class-1 Left-Behinds; only three are class-5 Winners.

Code	Language	Script	Family	Class	Inclusivity
as	Assamese	Bengali	Indo-European	2	✓
brx	Bodo	Devanagari	Sino-Tibetan	1	✗
bn	Bengali	Bengali	Indo-European	5	✓
doi	Dogri	Devanagari	Indo-European	1	✗
en	English	Latin	Germanic	5	✓
gom	Konkani	Devanagari	Indo-European	1	✗
gu	Gujarati	Gujarati	Indo-European	4	✓
hi	Hindi	Devanagari	Indo-European	5	✓
kha	Khasi	Latin	Austroasiatic	1	✗
kn	Kannada	Kannada	Dravidian	4	✓
ks	Kashmiri	Arabic	Indo-European	1	✗
mai	Maithili	Devanagari	Indo-European	1	✗
ml	Malayalam	Malayalam	Dravidian	4	✓
mni	Manipuri	Meithi	Sino-Tibetan	1	✗
mr	Marathi	Devanagari	Indo-European	4	✓
ne	Nepali	Devanagari	Indo-European	2	✗
or	Odia	Odia	Indo-European	3	✓
pa	Punjabi	Gurumukhi	Indo-European	3	✓
sa	Sanskrit	Devanagari	Indo-European	2	✗
sat	Santali	Ol Chiki	Austroasiatic	1	✗
sd	Sindhi	Arabic	Indo-European	1	✗
ta	Tamil	Tamil	Dravidian	4	✓
te	Telugu	Telugu	Dravidian	4	✓
ur	Urdu	Arabic	Indo-European	5	✓

The IndicXTREME task suite

9 tasks across 4 categories — 5 classification, 2 structure prediction, 1 QA, 1 retrieval. Since the benchmark evaluates **zero-shot** transfer, only test (and where possible, in-language dev) sets are built — models are always trained on English.

Task Category	Dataset	Task	Dev	Test	Method	Lang.	Metric	Domain
Classification	IndicSentiment	Sent. Classification	156	1000	HA	13	Acc.	Reviews
	IndicXNLI	NLI	2490	5010	MT [§]	12	Acc.	Misc
	IndicCOPA	Reasoning	-	500	HA	18	Acc.	Misc
	IndicXPara	Sent. Equivalence	-	2002	HA	10	Acc.	Misc.
	M-Intent	Intent	2033	2974	HA	7	Acc	Spoken
Structure Prediction	Naamapadam	NER	52-13460	607-1080	HA	9	F1	News
	M-SlotFill	Slot Filling	2033	2974	HA	7	F1	Spoken
QA	IndicQA	Span Extraction	-	1517-2017	HA	11	F1	Wiki.
Retrieval	FLORES	Sent. Retrieval	-	1012	HA	18	Acc.	Wiki++

HA = human annotated, **MT** = machine translated. Of 105 total evaluation sets summed across languages and tasks, 52 are brand-new contributions.

Building IndicCorp v2: finding sources

The corpus prioritizes clean, diverse text, so news articles are chosen as the primary source — a deliberate departure from generic CommonCrawl dumps like CC-100 and mC4.

DISCOVERING NEW SOURCES

Beyond the sources already known from IndicCorp v1, new ones are found via news repositories and automatic web search: the most frequent words in a language are used as search queries, and the language of the retrieved pages is inspected to identify candidate source URLs.

This process is noisy — some retrieved URLs turn out to host offensive or machine-generated content, which motivates the next step.

The quality gate: human-verified URLs

Before any crawling happens, every candidate source is checked by a person — this is the step that distinguishes IndicCorp from CC-100 and mC4 in Table 1's "Verified source URLs" row.

THE VERIFICATION TASK

Each annotator visits a candidate URL and confirms it is a genuine website containing clean data in the language of interest. Across languages, **1–33% of URLs are discarded** as noisy. Only the shortlisted, verified URLs are handed to the open-source `webcorpus` toolkit for crawling.

Notice this quality gate exists even for the *pretraining* corpus — human supervision in this work is not limited to the evaluation benchmark.

Post-processing the crawled dumps

Three automatic filters clean the crawled text before it enters the corpus:

Language ID (LID)

Paragraph-level filtering with cld3 and langdetect in parallel (cld3 misses Assamese & Oriya). Bodo and Dogri aren't supported by either tool, so LID filtering is skipped for them.

Script-based cleaning

Crawled sentences often mix in transliterations or English phrases. A sentence is dropped if the fraction of characters in its native script is **below 0.75**.

Punctuation-based cleaning

Punctuation is stripped from each sentence; if what remains is shorter than **10 words**, the document is discarded entirely.

Filtering offensive content

Web-crawled Indic text is known to contain offensive material (Kreutzer et al., 2022a). Two blacklists are merged to catch it:

IN-HOUSE BLACKLIST

In-house annotators curate offensive words and phrases for 17 languages — about 90 words/phrases per language, including borderline terms flagged by native speakers as offensive only in certain contexts.

NLLB BLACKLIST

A parallel, independently released list covering 209 languages (Costa-jussà et al., 2022) is merged in, widening coverage beyond the 17 in-house languages.

A sentence is removed if it contains a blacklisted word, or (for phrases) if the whole phrase appears verbatim. This single filter shrinks the corpus from **23.1B to 20.9B tokens** — a 9.5% reduction, confirmed by direct computation from the paper's own figures.

IndicCorp v2: the resulting scale

L	v1	v2	L	v1	v2
as	32.6	67	ml	721	931
brx	-	2.5	mni	-	0.6
bn	836	926	mr	551	795
doi	-	0.1	ne	-	852
en	1220	6501	or	107	122
gom	-	31.9	pa	773	732
gu	719	901	sa	-	125
hi	1860	6107	sat	-	4
kha	-	46	sd	-	13.2
kn	713	875	ta	582	476
ks	-	0.06	te	674	731
mai	-	13.7	ur	-	667
Total			8789	20920	

The de-duplicated v2 corpus totals **20.9B tokens** across 24 languages — **2.38×** the size of v1's 12-language, 8.79B-token corpus (paper rounds to "2.3×"), with Hindi alone growing **3.28×** ("3.3×" in the paper).

Many languages — Bodo, Dogri, Konkani, Kashmiri, Maithili, Manipuri, Sanskrit, Santali, Sindhi, Urdu — simply didn't exist in v1 at all (shown as "-").

The bottom 11 lowest-resource languages together still contribute only 1.08B tokens — a fraction of Hindi alone.

Creating IndicCOPA from scratch

COPA (Choice of Plausible Alternatives) tests causal commonsense reasoning: given a premise, pick which of two alternatives is the more plausible cause or effect.

TRANSLATION WITH A BIAS GUARD

The English COPA test set is manually translated into 18 Indic languages. Before translation, the premise and its two choices are **randomized and reassigned across translators** — so no single translator sees a whole item and could unconsciously preserve tells (like sentence length) that leak the answer. The three pieces are re-grouped into complete items only after translation.

Fine-tuning uses the English Social IQA dataset, since COPA itself has no training split.

Creating IndicQA: choosing the source articles

IndicQA is a cloze-style reading-comprehension dataset built from scratch, with no English-source translation involved anywhere in its creation.

1. Manually collect a list of topics tied to Indic history, monuments, authors, politicians, festivals, etc.
2. Rank topics by how many Indic-language Wikipedias carry an article on them.
3. Discard topics whose articles average fewer than 10 sentences.
4. Use the articles of the remaining, top-ranked topics as context paragraphs for QA pairs.

The result: 11 languages, contexts drawn from real encyclopedic text about Indic culture — not machine-translated from an English original.

IndicQA's trick: avoiding lexical overlap

Annotators asked to write a question directly from a passage tend to copy a sentence and prepend a "wh" word — producing questions that are trivial lexical matches. IndicQA's two-phase process breaks this shortcut:

PHASE 1 — WRITE THE QUESTION

Split each context into a larger first part and a smaller second part (2–3 sentences). Machine-translate *both* parts to English. An annotator writes Indic-language questions from the **English translation** — they cannot copy a sentence they never saw in Indic.

2–3 annotators per language; all annotation done on the Haystack tool.

PHASE 2 — MARK THE ANSWER

A *different* annotator sees only the original Indic first part and marks answer spans for the questions. Questions drawn from the withheld second part become naturally **unanswerable** — this is exactly where the NA labels come from.

IndicQA's annotation guidelines

QUESTION CREATION

- At least 2 questions per paragraph
- Answer span ≤ 5 continuous words
- Unambiguous even without the context
- Minimize phrase overlap with the paragraph
- Answer must lie in a single sentence

ANSWER MARKING

- Answer is a continuous span, ≤ 5 words
- Never an entire sentence
- Never a pronoun
- If the answer string appears more than once, mark the occurrence most relevant to the question

Rules this specific are what keep 11 independently-annotated languages comparable.

IndicQA by the numbers

L.	Q	A	NA	L.	Q	A	NA
as	1789	1225	564	mr or pa ta te			
bn	1763	1263	500		1604	1108	496
gu	2017	1273	744		1680	1279	401
hi	1547	1052	495		1542	1181	361
kn	1517	1138	379		1804	1276	527
ml	1589	1101	488		1734	1398	336
Total					18579	13283	5292

Across 11 languages: **18,579** questions, of which **13,283** (71%) are answerable and **5,292** (28%) are unanswerable — the direct product of the Phase 1 / Phase 2 split on the previous slides.

Creating IndicXParaphrase

1001 English sentences (≥ 10 words) are machine-translated into 10 languages with IndicTrans; annotators verify/correct each translation, then write **one paraphrase and one non-paraphrase** per sentence — a 1001-way parallel triplet in every language.

WRITING A PARAPHRASE

Minimize word overlap; swap active/passive voice; use synonyms liberally; use temporal phrase swapping, e.g. *"he fell and got hurt"* $\rightarrow$ *"he got hurt when he fell."*

WRITING A NON-PARAPHRASE

Swap named entities, pronouns, adjectives, adverbs, e.g. *"John drove Jane to the market"* $\rightarrow$ *"Jane drove John to the market."* Negation/antonyms restricted unless necessary.

Unlike PAWS-X, no back-translation or noisy alignment is used to manufacture non-paraphrases. 2 annotators per language; the task is run on Google Sheets.

Creating IndicSentiment

Ordinary product reviews are one-dimensional and highly polarized, which makes sentiment classification artificially easy. IndicSentiment is designed to be harder and more nuanced.

1. Curate products from 15 popular categories on Amazon, Flipkart, and Myntra.
2. Ask one set of annotators to list the aspects that matter for each product (e.g., for a dress: material, color, sleeves).
3. Ask a *different* set of annotators to write natural positive and negative reviews that concentrate on one or more of those specific aspects.
4. Reviews are written in English first, then manually translated into 13 Indic languages — a 13-way parallel dataset.

2 annotators per language; done on Google Sheets; annotators are instructed to avoid offensive language.

Adapting rather than creating: the other four datasets

Not every task in IndicXTREME is built from scratch — some adapt existing multilingual resources, still under human supervision:

IndicXNLI

Aggarwal et al.'s (2022) machine-translated XNLI. Source text is transcribed speech, so translations can be structurally odd — this work manually re-verifies and corrects parts of the test set (ongoing, see next slide).

Naamapadam (NER)

English sentence NER-tagged by an off-the-shelf model → tags projected to the Indic sentence via word alignment → annotators verify and correct the projected tags (IOB2 format).

FLORES (retrieval)

Already an n-way parallel, human-translated dataset (1012 sentences, 18 Indic languages) — incorporated directly for sentence-retrieval evaluation.

MASSIVE (intent / slot-filling)

Amazon Alexa user queries, human-annotated for 51 languages; a 7-Indic-language subset is taken as-is for IndicXTREME.

IndicXNLI: verification as an ongoing process

Lang.	Ver.	Corr.	Lang.	Ver.	Corr.
as	3000	1918	mr	1648	944
bn	1510	835	or	2107	1820
gu	-	-	pa	-	-
hi	4000	1142	ta	-	-
kn	1370	264	te	872	527
ml	3200	2427			

Lang.	Org.	HV*	Lang.	Org.	HV*
as	71.6	72.0	mr	73.2	73.5
bn	76.3	76.5	or	74.0	73.5
gu	75.6	75.6	pa	77.2	77.7
hi	77.5	77.5	ta	74.5	74.5
kn	74.7	74.7	te	75.2	75.0
ml	74.9	73.7			

Of 5,010 test instances per language, only a subset has been manually re-verified so far (e.g. 4,000 of Hindi's, 1,142 corrected). Scores on the partially-verified set (HV*) are already close to the original (Org.) — average unchanged at 75.0 — but the authors are explicit that this is a live, six-month effort, not a finished dataset.

The common thread: nothing is purely automatic

Whichever of the four creation paths a task followed — translated from scratch, built with no English source at all, machine-translated then verified, or tag-projected then corrected — every single evaluation set in IndicXTREME carries a human-supervision step:

DIRECTLY FROM THE PAPER

"We reemphasise that ALL the evaluation sets included in IndicXTREME were created with human supervision. In other words, they were either translated or post-edited or created or verified by humans."

Three tasks — NER, QA, and paraphrase detection — were created entirely from scratch, with **no translation from English sources** at any stage.

The evaluation protocol: zero-shot fine-tuning

IndicXTREME only ships test (and some dev) sets — never Indic-language training data. This forces every model into the same zero-shot recipe:

1. Fine-tune the pretrained LM independently for each task, using **English** training data.
2. Select the best configuration using the **English development** set.
3. Evaluate the chosen model directly on each Indic-language test set — no Indic data was ever seen during training or model selection.

Unlike most prior work, which reuses one fixed hyperparameter setting across all tasks, this paper runs a **task-specific** grid search (learning rate, weight decay) for every model, for a fair comparison.

Does an in-language dev set actually help?

Models	IndicSentiment		Naamapadam		MASSIVE (Intent)		IndicXNLI	
	in-lg.	in-fam.	in-lg.	in-fam.	in-lg.	in-fam.	in-lg.	in-fam.
mBERT	72.9 _{+3.4}	72.9 _{+3.4}	65.8 _{+2.8}	65.2 _{+2.3}	15.1 _{+1.9}	14.7 _{+1.5}	58.4 _{+3.7}	58.4 _{+3.7}
XLMR	86.1 _{+2.1}	84.6 _{+0.6}	73.0 _{+1.3}	73.0 _{+1.3}	67.6 _{+1.0}	67.6 _{+1.0}	70.4 _{+0.7}	70.1 _{+0.4}
MuRIL	89.3 _{+4.2}	89.2 _{+4.1}	74.3 _{+0.0}	74.1 _{−0.2}	77.3 _{+0.1}	77.5 _{+0.3}	74.0 _{+1.6}	74.0 _{+1.6}
IndicBERT	92.5 _{+4.2}	92.5 _{+4.2}	73.2 _{+0.0}	73.2 _{+0.0}	79.1 _{+0.3}	79.1 _{+0.3}	73.0 _{+0.0}	72.6 _{+0.4}
+Samanantar	92.4 _{+4.1}	92.4 _{+4.1}	72.9 _{+0.5}	72.9 _{+0.5}	79.2 _{+0.4}	78.9 _{+0.1}	74.3 _{+0.0}	74.3 _{+0.0}
+Back-Trans.	93.1 _{+5.6}	92.8 _{+5.3}	72.2 _{+0.4}	72.2 _{+0.4}	77.5 _{+0.1}	77.4 _{+0.0}	71.5 _{+0.8}	71.5 _{+0.8}

Every subscript is a gain over using an English dev set. Hindi and Tamil dev sets stand in for the whole Indo-European and Dravidian families respectively — and models selected with these **in-family** sets perform on par with those selected using true **in-language** sets, both clearly ahead of English-only selection.

Hindi is a better transfer language than English

Models	Indic Sentiment	Indic XNLI	Indic COPA	Indic XPara.	MASSIVE (Intent)	Naama- Padam	MASSIVE (Slotfill)	Indic QA
IndicBERT v2 +Samanantar	88.3	74.3	63.0	57.0	78.8	72.4	57.3	49.2
gold zero-shot	-	-	-	-	81.9	75.9	67.9	-
silver zero-shot	90.3	77.0	51.9	57.5	-	-	-	46.4

Swapping the transfer language from English to Hindi ("gold zero-shot," using Naamapadam and MASSIVE's real Hindi training data) lifts NER by **+3.5**, Intent by **+3.1**, and Slot-filling by **+10.6** points. "Silver zero-shot" (machine-translating the English training set to Hindi) helps sentiment/XNLI/paraphrase similarly, but *hurts* IndicCOPA and IndicQA — both tasks are unusually sensitive to the small errors translation introduces.

Does the benchmark distinguish good models? Yes.

Models	Classification					Structure Prediction		QA	Retreival
	Indic Sentiment	Indic XNLI	Indic COPA	Indic XPara.	MASSIVE (Intent)	Naama- Padam	MASSIVE (Slotfill)	Indic QA	FLORES
IndicBERT v1	61.8	42.8	51.0	47.5	-	25.3	-	10.1	1.1
mBERT	69.5	54.7	51.7	55.2	13.2	63.0	6.2	32.9	32.3
XLMR	84.0	69.7	60.1	56.7	66.6	71.7	50.0	44.8	3.1
MuRIL	85.1	72.4	58.9	60.8	77.2	74.3	57.0	48.3	52.3
v1-data	85.7	66.4	52.4	49.6	25.8	58.3	34.4	37.6	54.9
IndicBERT v2	88.3	73.0	62.7	56.9	78.8	73.2	56.7	47.7	69.4
+Samanantar	88.3	74.3	63.0	57.0	78.8	72.4	57.3	49.2	64.7
+Back-Trans.	87.5	69.7	53.8	50.7	77.4	71.9	54.6	42.2	68.6
IndicBERT-SS	88.1	73.9	64.2	56.4	80.7	66.6	57.3	49.7	71.2

The IndicBERT v2 family beats every massively-multilingual baseline (mBERT, XLM-R, MuRIL) on **7 of 9 tasks**. The pattern holds even for the ablation trained purely on IndicCorp v2 with MLM (no parallel data) — models that don't have to share capacity with unrelated high-resource languages simply do better on Indic tasks.

The gap the benchmark exposes: Winners vs. Left-Behinds

	as	bd	bn	gom	gu	hi	kn	ks	ml	mai	mr
mBERT	38.8	49.5	43.5	50.2	51.5	47.6	42.1	11.9	39.5	43.5	53.2
XLMR	48.2	51.6	53.3	56.4	55.2	58.2	52.2	0.3	54.1	28.7	52.9
MuRIL	60.3	48.8	67.2	52.0	67.7	68.4	67.8	9.3	66.9	36.8	66.3
v1-data	61.8	60.2	57.2	47.8	63.6	58.1	55.6	0.6	54.9	35.5	61.9
IndicBERT	68.4	80.4	69.1	58.2	70.0	68.6	67.5	0.9	67.9	49.7	67.4
+Samanantar	68.0	87.8	69.2	58.2	69.8	69.1	68.2	0.4	68.1	46.2	67.7
+Back-Trans	63.0	82.7	66.7	51.8	64.8	66.9	65.4	1.8	64.9	47.8	64.5
IndicBERT-SS	69.1	89.7	69.4	63.4	68.8	68.2	67.8	2.1	68.6	70.5	66.9
	mni	or	pa	sa	sat	sd	ta	te	ur	avg	
mBERT	2.5	31.4	51.9	30.9	26.3	50.6	40.0	43.7	44.4	39.6	
XLMR	0.3	52.4	51.9	28.0	26.9	64.0	53.4	56.4	54.0	44.9	
MuRIL	0.7	60.5	64.9	46.4	25.1	58.0	66.2	64.2	68.2	53.3	
v1-data	1.9	63.5	63.4	28.2	25.6	53.0	53.6	58.0	24.0	46.4	
IndicBERT	0.3	70.2	68.9	44.0	34.0	59.2	68.0	70.2	72.0	57.8	
+Samanantar	0.3	70.0	69.2	37.3	28.3	52.4	68.3	70.6	71.2	57.0	
+Back-Trans	0.5	65.0	65.9	46.4	27.9	51.0	65.8	66.9	68.2	54.9	
IndicBERT-SS	0.2	71.5	68.4	47.1	37.5	63.4	68.3	70.1	74.8	60.3	

Extremely low-resource languages (below the 10th percentile of Table 3) show a sharp, consistent drop — e.g. on IndicCOPA, Santhali and Sindhi score 25.9% and 17.7% lower than Hindi respectively. The paper attributes this to lacking pretraining data *plus* two compounding effects: no shared script, and no linguistic "cousin" in the corpus to bridge transfer. And even this is optimistic — IndicXTREME can only evaluate 19 of IndicCorp's 24 languages; five "left-behind" languages have no evaluation set at all yet.

Ethics and annotator practices

ANNOTATORS

Paid a competitive monthly salary set by qualification, prior experience, and local government norms. All were native speakers of the language they worked on, from the Indian subcontinent, and were informed their annotations would be publicly released.

Code and models are released under the MIT License; the datasets under CC-0.

DATA

No personally identifying information in any annotated set. Both the annotated data and the crawled corpus were checked for offensive content and discarded where found.

Reading list

- [Doddapaneni et al., *Towards Leaving No Indic Language Behind*](#) — ACL 2023, the source for every table in this deck.
- [Joshi, Santy, Budhiraja, Bali & Choudhury, *The State and Fate of Linguistic Diversity and Inclusion in the NLP World*](#) — the "Winners" / "Left-Behinds" taxonomy.
- [Hu et al., *XTREME*](#) and [Ruder et al., *XTREME-R*](#) — the multilingual benchmarks IndicXTREME responds to.
- [Aggarwal, Gupta & Kunchukuttan, *IndicXNLI*](#) — the machine-translated NLI set being manually re-verified.
- [Mhaske et al., *Naamapadam*](#) — the NER dataset built via tag projection.
- [Costa-jussà et al., *No Language Left Behind*](#) — source of the 209-language offensive-word list merged into IndicCorp's filter.
- [FitzGerald et al., *MASSIVE*](#) — source of the M-Intent and M-SlotFill tasks.
- [Official IndicBERT code, data, and model checkpoints.](#)